

Heidi Pamela Martínez Martínez
1952947 N1

Actividad 3.4

$K=3$
 $n=12$

Máquina	1	Operador 2	3	Σ
1	42.5	34.3	39.6	121.4
2	39.8	40.1	40.5	120.4
3	40.2	40.5	41.3	122.0
4	41.3	42.2	43.5	127.0
Σ	163.8	162.1	164.9	490.8

$H_0 = \alpha_1 = \alpha_2 = \alpha_3 = \alpha_4$
 $H_a = \text{Al menos 1 } \alpha \text{ es diferente}$

$$f_T = \frac{2.884}{1.432} = 2.014$$

$$SCE = 18.24 - 8.653 - 0.995 = 8.592$$

$$f_B = \frac{0.497}{1.432} = 0.347$$

$$C = \frac{490.8^2}{12} = 20\,073.72$$

$$STC = (42.5^2 + 39.8^2 + 40.2^2 + 41.3^2 + 34.3^2 + 40.1^2 + 40.5^2 + 42.2^2 + 39.6^2 + 40.5^2 + 41.3^2 + 43.5^2) - 20\,073.72 = 18.24$$

$$SCB = \left(\frac{163.8^2}{4} + \frac{162.1^2}{4} + \frac{164.9^2}{4} \right) - 20\,073.72 = 0.995$$

$$SCT = \left(\frac{121.4^2}{3} + \frac{120.4^2}{3} + \frac{122.0^2}{3} + \frac{127.0^2}{3} \right) - 20\,073.72 = 8.653$$

$$CMB = \frac{0.995}{2} = 0.497 \quad CMT = \frac{8.653}{3} = 2.884 \quad CME = \frac{8.592}{6} = 1.432$$

Var	Suma de cuadrados	Grados de libertad	Cuadrados medios	f
Máquinas	8.653	3	2.884	2.014
Operadores	0.995	2	0.497	0.347
Error	8.592	6		
Total	18.24	11		

Respuesta ↴

$$F_c = 2.014 < f_t = 4.76$$

No se rechaza H_0 ya que existe la existencia de la igualdad de las medias